

Nitin Sunil Chaube

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PROFESSIONAL SUMMARY

AI/ML Engineer with 3 years of experience building and deploying production-grade machine learning systems that directly improve business outcomes, driving measurable gains in model accuracy, operational efficiency, and real-time predictive performance at enterprise scale.

EDUCATION

Stevens Institute of Technology | *M.S., Computer Science* (GPA: 4.0)

Hoboken, New Jersey Sep 2024 - May 2026

Mumbai University | *B.E., Computer Science* (GPA: 8.8)

Mumbai, India Aug 2018 - May 2022

EXPERIENCE

InterviewReady.io | *AI Research Engineer*

Jan 2026 - Mar 2026

- Researched and implemented SOTA agentic AI patterns, including ReAct, Plan-and-Solve, and Reflection, to create engineering-grade reference architectures for 20,000+ technical learners, with emphasis on stateful memory persistence and safe tool calling.
- Investigated multi-node deep learning training protocols using PyTorch FSDP and DDP, along with long-context inference techniques, translating research papers into production-ready Python implementations for latency-sensitive applications.
- Designed end-to-end coding labs for "Building Production Agents," creating reproducible environments for testing multi-agent collaboration, hallucination mitigation, and structured output validation.

Teacher Insurance and Annuity Association of America (TIAA) | *ML Engineer*

Jul 2022 - Aug 2024

- Architected and deployed end-to-end machine learning data pipelines on AWS for NLP-based financial incident triage, processing 30,000+ monthly logs with 94% accuracy by fine-tuning BERT embeddings on a domain-specific financial lexicon and reducing regulatory reporting risks.
- Engineered an intelligent incident routing system that merged ML predictions with deterministic business rules. This hybrid approach reduced manual operational workload by 70% (saving 100+ hours/month) while maintaining strict audit compliance for high-priority workflows.
- Built scalable, fault-tolerant microservices and asynchronous data pipelines handling 1,000+ daily incidents with 99.9% uptime, leveraging Apache Kafka to ensure zero data loss and eventual consistency across federated banking systems.
- Proposed real-time model drift detection and predictive monitoring using Splunk and Dynatrace, enabling the infrastructure team to preemptively resolve 500+ critical production issues annually before they impacted customers.

Stevens Institute of Technology | *Teaching Assistant for CS 513 and CS 284*

Jan 2025 - May 2025

- CS 513: Formulated end-to-end ML projects on real-world datasets using Python, enforcing advanced feature engineering, hyperparameter tuning, cross-validation, and interpretability; mentored the class of 70 students to debug pipelines, resolve data leakage/imbalance/overfitting.
- CS 284: Data Structures: Led labs and 1:1 debugging for 50+ undergraduates on graphs, hashing, recursion, balanced trees, and DP; collaborated on coursework to translate algorithms into scalable, optimized code with Big-O analysis.

PROJECTS

Advisor AI | [Github](#) | <https://advisoraii.web.app/>

- Architected a multi-agent LangGraph StateGraph pipeline that orchestrated parallel agents, accelerated query latency by 65% compared to sequential execution, and achieved sub-2-second response times under heavy concurrent load.
- Designed an entity-aware hybrid retrieval system merging regex-based detection (80+ course prefixes, faculty matching) with semantic search (BAAI/bge-small-en-v1.5) and BM25 scoring; achieved 96% retrieval precision across a 4-collection ChromaDB index of 87,000+ domain records via metadata-filtered routing.
- Implemented multi-layer AI safety guardrails for prompt-injection detection, LLM-based classification, and identity protection, along with a self-reflection quality gate that automatically refined responses scoring below 7/10; reduced hallucination rates to less than 3% across 10,000+ production queries.
- Built a synthetic-data distillation pipeline: prompted Gemini to generate 87K+ high-quality domain Q&A pairs from raw Stevens corpus chunks, then fine-tuned Qwen2.5-14B-Instruct on the distilled dataset, pivoting after an initial QLoRA-tuned LLaMA-2-7B baseline underperformed on domain accuracy; applied AWQ 4-bit post-training quantization (28GB to 9GB) and served via vLLM in Docker/Kubernetes, reaching 92–94% domain accuracy at 250–400ms time-to-first-token.

Conversation Intelligence Platform | [Github](#)

- Architected an end-to-end Azure AI contact-center analytics platform featuring a RAG chatbot, compliance auditor, structured extraction, and human-in-the-loop QA by extending Microsoft's Conversation Knowledge Mining accelerator with an end-to-end voice streaming pipeline.
- Engineered an Azure OpenAI compliance auditor that scores every call against 6 severity-weighted policy rules (PII, greeting, escalation, resolution, empathy, upsell) with inverted-logic handling and retry/backoff for rate limits replacing less than 2% manual QA sampling with 100% automated coverage.
- Built a GPT-4o-mini structured extraction pipeline that derived 10+ operational metrics per call, including sentiment, topic, intent, resolution, handle time, customer effort score, and hold/transfer detection, then persisted the results to Azure SQL and Azure AI Search as a medallion data layer for downstream BI.
- Developed a 6-module operations dashboard (Operations, Agent Performance, Compliance Monitor, QA Review, AI Accuracy, Conversation Explorer) and an Azure SQL-backed Power BI executive dashboard, both consuming the same data layer to serve supervisors and executives from one pipeline.

SKILLS

- **Programming:** Python, JavaScript, SQL, C++, Bash, PyTorch, TensorFlow, JAX, ADK, Hugging Face Transformers, Pandas, NumPy
- **GenAI & Agentic AI:** GPT, Gemini, Claude, LLaMA-2, BERT, BGE/Sentence-Transformers, LangGraph, LangChain, AutoGen, QLoRA/LoRA Fine-Tuning, RLHF, Inference Optimization (Quantization, KV-Cache), ReAct, Plan-and-Solve, Reflection, Tool Calling, Function Execution, Multi-Agent Orchestration
- **RAG & Retrieval:** Vector Databases (ChromaDB, Pinecone, FAISS), Hybrid Search, BM25, Re-ranking, Hallucination Guardrails, Evaluation (Ragas, TruLens)
- **Cloud & MLOps:** AWS, GCP (Cloud Run, Vertex AI), Docker, Kubernetes, CI/CD, MLflow, Weights & Biases, Model Monitoring, A/B Testing, Splunk, Dynatrace, Git
- **Data & Backend:** FastAPI, Flask, MongoDB, PostgreSQL, Redis, Apache Kafka, Spark, Celery, Async I/O, Microservices, SSE/WebSocket Streaming

PUBLICATIONS / CERTIFICATIONS

- DIG (Disaster Information Graph). *Published a comprehensive disaster response framework integrating high performance NLP classification (94.2% accuracy), temporal Knowledge Graphs for RAG-based event clustering, and real-time streaming analytics to improve situational awareness and emergency response efficiency.* (- <https://tjjer.org/tjjer/papers/TJJer2309075.pdf>)
- Microsoft Certified: Azure AI Engineer Associate.